

Menstrual Monster: A Tangible Interactive Co-educational Game Designed for Teenagers

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ABSTRACT

Learning menstruation in early adolescence could reduce teenagers' misunderstanding of it and help them treat menstruation in a proper way. This paper explored a tangible game for teenagers of different genders learning menstruation through collaborative playing. The game included five levels where users play together and learn the cause, products, symptoms of menstruation as well as try to judge some scenarios and listen to audios about menstruation. In our user study, we invited three groups of teenagers ages 11 to 16. Each group contained at least one male and one female, and we let them play the game freely. Teenagers were successfully able to play the game collaboratively, learn menstruation-related knowledge. The results revealed motivation differences related to gender, and after the game, teenagers demonstrated the observable change of the attitude towards menstruation.

CCS CONCEPTS

• **Human-centered computing**; • **Interaction design**; • **Empirical studies in interaction design**;

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KEYWORDS

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1 INTRODUCTION

Menstrual shame continues to persist worldwide. Stigmas about menstruation and body fluids drive menstruation as a cultural taboo, which leads to a misunderstanding of women's physiology and shame [1, 2]. Previous research has focused on helping women develop a proper understanding of their own bodies, and reduce their shame on menstruation. In fact, menstrual shame from a male perspective is also an important cause of menstrual stigma. Men's lack of understanding of menstruation can exacerbate the intimate nature of menstrual topics [10]. We conducted a study to explore the possibility of joint gender participation in menstrual education. To promote participation, we adopted the medium of games. The study was conducted with the following general objectives: 1) to understand adolescents' experiences, knowledge, and attitudes related to menstruation, 2) to carry out the popularization of menstruation-related knowledge through tangible games, and 3) to explore the effects of tangible interactive co-education games on adolescents' attitudes toward menstruation.

In the game design, we considered three main design principles: 1) create meaningful game goals [3,6], 2) create and embed an imaginary context [7], and 3) provide cognitive challenges [6]. Based on the design principles, we provided an entire story context for the game: fixing the 'menstrual monster'. After finishing each level, participants would get one clue card, which could be used to exchange one questionable card covered on 'menstruation monster'. The mechanism of exchanging cards represented getting rid of bias and understanding menstruation. Besides, this mechanism could keep users engaged. The participant sample consisted of eight adolescents in the age group of 11-16 years old, and the grouping in the experiment was two groups of siblings and one group of classmates. Semi-structured interviews were conducted with each participant before and after the game to understand their attitudes towards menstruation and their experience of the "menstrual monster" game. Participants demonstrated motivation differences related to genders during the game. In addition, participants generally showed more positive attitudes and curiosity about menstruation later in the game. We describe these changes and discuss the reasons in this paper. This project builds on previous work to explore the impact of the Co-educational Game on adolescent menstrual attitudes through tangible interactions. The aim is to teach boys and girls menstruation knowledge and proper attitudes, generate open discussion on the topic of menstruation, and try to help them get rid of menstrual shame.

2 RELATED WORK

2.1 Menstrual education

The first menstruation, which usually occurs between the ages of 9 and 15, is a key event in female puberty and is associated with other physical and psychological changes during puberty [1]. In a study of menstrual attitudes by McPherson [4], it was shown that early menstruation can have a lasting effect on women's menstrual attitudes and perceptions of their own bodies. Therefore, timely and adequate menstrual education is necessary for the healthy physical and mental development of women. In recent years, the topic of menstruation has received more and more public focus [3, 8, 9] TULI [8] explored the sociotechnical implications for the design of menstrual health education initiatives through the lens of feminist HCI. "Curious Cycles" was designed to invite menstruating women to experiment and explore their bodies, developing participants' appreciation of their own bodies and rebelling against the stigmatization of menstruation and bodily fluids [1]. However, most of these designs and studies focus on women's menstrual experiences, reflecting a current state of menstrual education - there are clear gender differences. In a study by Kalman on menstruation and single-parent families, it was shown that in the United States, there is a growing number of families consisting of single fathers and their children, where it is important for single fathers to act as life coaches to acknowledge the physical changes in adolescent girls to reduce anxiety from menarche. But the fact is that fathers are generally unprepared for these issues, and girls find it embarrassing to discuss menstruation with men [10]. Men are less often included in menstrual education, and this serious gender disparity in menstrual education can promote the private nature of menstrual topics and

exacerbate cultural taboos. It also leads some men to neglect their involvement in menstrual education when they become elders.

2.2 Tangible interaction

Tangible interactions have many benefits in supporting or improving children's learning [11–14, 28]. Klemmer suggested that physical objects through physical contact can facilitate active learning and improve understanding [15]. Similarly, Hornecker [16] believed that touchable objects can enhance sensory pleasure and enjoyment by awakening the sense of touch, which promotes users' exploration. Horn, Crouser, and Bers [17], through a review of three tangible computer programming studies, found that tangible interactions are more likely to trigger positive cooperation and have less obvious gender differences. Jamal [18] developed a game to assess levels of socioeconomic and gender biases among children, as well as to gauge its comparative benefits over traditional detection methods, confirming the potential of introducing social issues into tangible interactions. We use tangible interaction to design for menstruation. On the one hand, it allows users to recognize menstruation more directly through physical interaction. On the other hand, the exploratory of tangible interaction is beneficial to trigger discussion and active learning [15, 16, 19].

2.3 Game-based learning

Gamified design has been confirmed more beneficial compared to traditional modes of teaching knowledge [5, 20, 21, 29]. Many studies on menstrual education have used the medium of games to reduce the seriousness of this topic [22] which remains taboos around the world. Menstrual Maze was designed for the community with the hope of exploring the possibilities of early menstruation education through parent-child play [23]. Jain and Yammiyavar [3] designed a collaborative learning game to alleviate taboos related to menstruation and puberty for adolescent girls in rural Assam. Referring to studies of other researchers, we attempt to understand the knowledge level and the attitudes of adolescents toward menstruation through a gamified design. In this design, we aim to promote discussion of menstruation by engaging boys and girls together.

3 MENSTRUAL MONSTER

3.1 Design principles

Game incentives could be used for reinforcement against prejudices [18]. Based on this conclusion, we propose three design principles to guide the design of the specific game mechanisms.

3.1.1 Create Meaningful Game Goals. In game-based cognitive learning, the outcome of the game loop contributes to learning, that is, when the player is motivated by the achievement of a specific achievable goal, the outcome produced by user feedback is the expected learning effect [3, 5]. Therefore, to make the learning of menstrual education more effective, we need to design a final goal for the game. The goal should be clear, achievable, able to challenge the player and make players be patient until the game is completed.

3.1.2 Create and Embed an Imaginary Context. The "context-availability" effects have been proven to be effective [6]. Contextualization brings players a specific problem situation, and the inter-relationship of knowledge and context helps players understand



Figure 1: Components and Userflow of ‘Menstrual Monster’. (a) 5 cards with questions on them, (b) 5 levels corresponding to question cards, (c) 5 clue cards, (d) menstrual Monster covered with 5 question cards, (e) take one question card before each level, (f) get one clue card for the level finished, (g) put the clue card on the corresponding position, (h) finish 5 levels in sequence and get Menstrual Monster restored.

the problem more easily, which is also supported by the shared cognition theory [25]. As a co-educational game, the knowledge should be introduced combining with a relevant scenario. With this in mind, we try to embed a meaningful and appealing fantasy context [24] that promotes engagement by arousing the player’s curiosity. This context should be consistent with the created game goal, that is, the context needs to act as a background of the game so that the goal of the game will no longer lack support.

3.1.3 Provide Cognitive Challenges. Stimulating cognitive curiosity is one of the characteristics of educational games. Cognitive challenges prompt players to solve puzzles. Since puzzles that are too simple may lose their challenges, the challenge should be interesting and unpredictable [5], requiring players to synthesize knowledge or skills. This process may also facilitate the introduction of new knowledge. In other words, our game should allow players to learn what they need to know in the cognitive challenges. Based on this principle, we hope that the content of menstruation can be integrated into the game, not simply by one-way transmission, but let the player accept the challenge in levels of our game, and make the cognitive behavior active.

3.2 Game design

‘Menstrual Monster’ is a tangible game for teenagers. The prototype of the monster is a sanitary napkin which satirizes the demonization of menstruation. Each player needs to finish five levels of the game in sequence (see figure 1), and they are allowed to play with peers for discussion. Before this game, five cards with questions on them are attached to the monster. Each question card is a metaphor of one stigma about menstruation and corresponds to one game level. Players need to take the question card of the level, for each level finished, they can get a clue card that painted part of the monster. The clue card should be placed on the monster instead of the corresponding question card,

thus recovering a part of the monster. After getting all 5 clue cards, players can restore the true appearance of the monster, which is cute, expressing that menstruation is perfectly normal and represents the beauty and health of a female’s body. Games can provide new perspectives for tabooed problems or issues that are rarely discussed in public [1]. We aim to use Menstrual Monster as a medium to introduce menstruation-related knowledge and help teenagers, both female and male, to learn and rethink menstruation.

3.2.1 Why Does Menstruation Occur. The first step of dispelling misunderstanding of menstruation is that understanding menstruation cycle is normal. In school, some physiological knowledge, including the cause of menarche would be taught to students who are in early adolescence. In order to help users adapt to the game, the cause of menstruation is designed as the content of our first game level. Inspired by Tran and Choi [23], as shown in figure 2, we use an image of the uterus, of which the key parts were labeled and explained. A small ball with conductive properties represents the egg cell. The users need to move the ball from the ovary to the vaginal outlet. When the ball is connected to the circuit, the LED matrix below will glow to present the effect of drops of blood, which symbolizes the arrival of menarche.

3.2.2 How to Use These. The second level is designed to introduce menstrual products, including menstrual pads, menstrual cups and tampons. In order to match the theme of the game and make it easier to identify [26], a uterus model with underwear and some menstrual products prototypes are used as the game prop (see figure 2). The surface of the prop slot was printed with detailed descriptions of the menstrual products, users could use these products model to learn how to use menstrual products, which reduces anxiety. Another important aim of this level is helping people learn more about menstrual cups and tampons because many females believe that the put-in menstrual products will destroy women’s chastity, which leads to women’s misunderstanding of their own bodies [2].

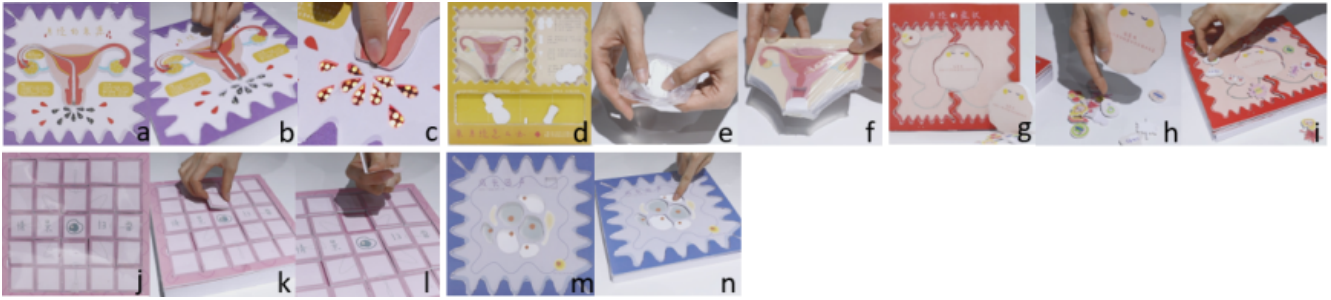


Figure 2: Userflow of 5 levels. (a) interface of level 1, (b) move the ball from the ovary to the vaginal outlet, (c) LED matrix below glow, (d) interface of level 2, (e) putting a sanitary napkin on the underwear, (f) put on the underwear, (g) interface of level 3, (h) take the symptom labels out of the box, (i) place the labels in the correct position, (j) interface of level 4, (k) pick up the scene cards, (l) read the contents, (m) interface of level 5, (n) push the button to play the audio.

3.2.3 Is This Normal. Many adolescent girls are worried about some menstrual symptoms that seem different from usual situations, some of these symptoms are normal while others are unusual. Girls tend to be unwilling to ask parents for help, which may do damage to their health. As shown in figure 2, in this level, we use different labels to represent various symptoms emerging before or during menstruation. Players need to judge whether the symptoms are normal or not. There were 7 normal and 7 unusual labels, the labels representing normal symptoms are coated with conductive ink [27] on the back. If all the labels are classified correctly, the circuit will be connected to turn the led on. For the unusual symptoms, we would tell the participants the reasons for these symptoms and suggest them going to the doctor.

3.2.4 Would It Be Embarrassing. Excessive privacy of menstruation leads to ignorance and stigma about menstruation. We believe the topic of menstruation should not just be among women, and boys need to have a better understanding of menstruation as well as other adults. At the same time, there should be a dividing line between privacy and knowledge about menstruation. We deploy the form of a minesweeping game. Various phenomena, behaviors, and thoughts about menstruation, such as putting the menstrual products in a dark-colored shopping bag because of embarrassment or thinking that menstrual blood is dirty, are written in the scenario cards. There are 24 scenario cards in the game, 6 of which with wrong or inappropriate scenarios and stands for landmines. When the players pick the card representing landmine, the light-sensitive sensors in the card slots will trigger the buzzer, and teenagers can more easily accept and change their attitudes in a familiar and specific context [24].

3.2.5 What about These Voices. At this level, we prepare audios related to menstruation, such as some comments with misunderstanding on menstruation, sounds of using menstrual products. For girls who have not experienced menstruation or adolescent boys, these sounds can arouse their curiosity. While for girls who have already experienced menarche, they may feel more familiar with these audios and may have more insight when hearing some of the ideas about menstruation. In addition to listening to the audio by pushing the buttons, players can also record their voices by pressing the record button. These voices will be uploaded to an empty

memory card prepared for the level as the level's audio material and heard by others. We hope that the positive or negative messages behind these voices will arouse thinking and promote discussion of menstruation among teenagers.

4 USER STUDY

4.1 Participants

We recruited three groups of participants in a shopping mall, as shown in table 1. These groups had different relationships and backgrounds. Before our experiment, we obtained parental consent of these participants.

4.2 Procedure

Before the experiment, we asked all the participants about their attitude and knowledge of menstruation. In addition, female participants were asked about their own menstruation experiences. Before playing the game, we introduced every level and told participants of each group that they should finish the game as a team. Each group was given 20 minutes to play the game. During the experiment, participants were encouraged to discuss the game content and finish the game together. After finishing the game, we conducted semi-structured interviews with each group, in which participants were interviewed together. Meanwhile, we asked them to provide feedback on the game. Finally, all participants were compensated with a souvenir worth \$5.

We obtained a total of 40 min of audio and 90 min of video in this experiment. For video data, we first sorted out the corresponding video data of each participant. We then watched the video and intercepted the keyframes. Then, we coded our data for emotion expression, partner interaction, game operation, and other obvious behaviors. For audio data, we translated and proofread the recordings of the participants. We also coded them into daily content, attitude expression, and experimental comment. Four research assistants participated in these works.

4.3 Results

4.3.1 Ease of Playing and Learning from the Game. Overall, participants were highly interested in playing and learning from Menstrual Monster. All participants told us they were having fun playing the

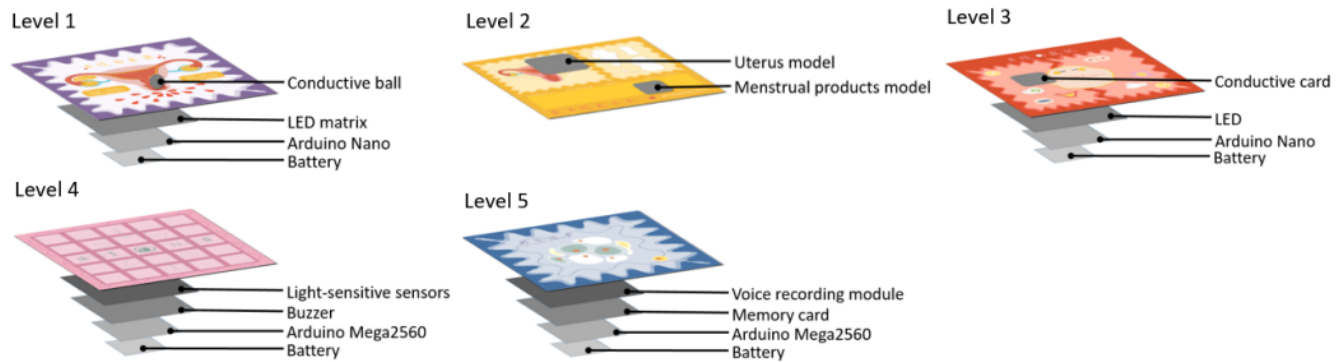


Figure 3: Implementation of Menstrual Monster.

Table 1: Participant Information

Group	Group 1	Group 2	Group 3
Participant	P1 & P2 (Sister & Brother)	P3 & P4 (Sister & Brother)	P5 & P6 & P7 & P8 (Classmates)
Age	16 & 11	13 & 11	14
Gender	Female & Male	Female & Male	Female & Male & Male & Male



Figure 4: Participants were interacting with Menstrual Monster.

game. With the game continuing, participants tended to be more engaged in it. This increased engagement was promoted by the novel interaction of our game. For example, when playing *would it be embarrassing* level, participants were attracted by the buzzing sound of picking “landmine” cards. P4 said: “I am trying to avoid triggering buzzer, which is really interesting”. After finishing the game, all participants mentioned that they learned something new through play. One male participant (P6) said: “Before playing the game, I didn’t

know that period would last several days”. Female participants also pointed out that many menstruation symptoms contained in the game were new to them, such as metrorrhagia and edema. One participant (P2) noted the advantage of learning with Menstrual Monster: “If this game could be used for teaching in school, we would be more interested in learning knowledge of menstruation. I think it also makes menstruation-related knowledge easier to understand for us teenagers”. As a bystander of our experiment, the

mother of teenagers in Group 2 said: “To be honest, it’s difficult for me to find an appropriate way to teach menstruation-related knowledge to children. But this game provides a good choice”.

4.3.2 Changing Attitudes before and after the Game. Before and after the game, generally, there were observable changes in the attitudes of participants, especially male participants. Most of the male participants (P2, P6, P7, P8) were gradually able to talk about menstruation openly. For instance, P2 mentioned that he was kind of awkward at the beginning of the game. When we asked for the reason, he said: “Because I am a boy, and I don’t have to learn this”. After finishing the game, he told us that he would take care of his sister when menstruation brought inconvenience to her. Similarly, four classmates of Group 3 were embarrassed at the start of the experiment. In *how to use these levels*, two male participants (P6, P7) even made fun of each other by choosing menstrual pads. But during the game, they gradually paid more attention to the game content including menstruation symptoms, people’s misunderstanding of menstruation. In the interview, they told us that having menstruation is perfectly normal, which should not be ashamed of.

4.3.3 Motivations and Concerns of Different Gender. During the game process, we noticed that motivations and concerns varied from gender. Female participants tended to be motivated by the connection between the game and their own experiences. In *Is this normal level*, they would compare some menstrual symptoms with their own experiences to judge the category of these symptoms, which helped them engage in the game. Furthermore, female participants concentrated more on the content of our game. Generally, all of them mentioned that some new knowledge was acquired from the game, which might be helpful for their life. For example, participants said that our game helped them know tampons and menstrual cups, and they would search for more details when needed. Besides, one participant (P6) expressed her expectation of including more content in the game, such as some tips for menstrual hygiene.

While the male participants were more likely to be motivated by the novel game mechanisms and playability. For example, after finishing some levels, P8 said: “I can’t wait to know what’s the next level would be like”. It should be noted that this motivation promoted their learning willingness. In specific, the results showed that *would it be embarrassing* was generally the most popular level among male participants. Three male participants (Group 3) competed with others for picking fewer cards that stand for wrong or inappropriate scenes. And they tended to read every card even though it was not picked by themselves. P8 said: “Competing with others makes me more eager to learn the knowledge”.

4.3.4 Promoting Discussion. Our experiment results indicated that Menstrual Monster could promote participants’ discussion. We noticed that all the male participants would tend to ask female participants for help when they encountered some questions they didn’t know, which stimulated discussion between them. For example, in order to finish *Is this normal level*, male participants would ask female participants whether they had similar symptoms or experiences. While in *What will everyone think level*, participants would discuss on audios and guess what it is together. When hearing the audio of tearing a sanitary napkin package, participants in Group 3 were confused. After discussion, they realized that they could

combine the sound with prior game levels and finally gave the right answer.

5 CONCLUSION AND FUTURE DESIGN

In this paper, we designed and developed ‘Menstruation Monster’, a collaborative menstrual educational game for teenagers. Through an empirical study, participants of the opposite sex experienced the menstrual monster together and shared their co-experiences of menstruation. We demonstrate the effectiveness of Menstruation Monster for adolescent menstruation education and point to the unique potential of cross-gender experiences in menstruation education.

In the future, we will expand the game content, for example, we will clarify more on the relation of gender and menstrual cycle for not all women menstruate and not all people who menstruate are women. We will also deepen in one modality in the interaction of sound and light, and engage in a longer study on how these specific interactions are meaningful. In addition, we plan to deploy ‘Menstrual Monster’ in elementary and middle schools, and pre- and post-tests will be carried out to verify the validity of our design.

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